Chapter 1

Introduction

The main purpose of these notes is to provide a detailed exposition of Abstract Algebra, Galois Theory and Algebraic Varieties over non-algebraically closed fields, with particular interest in the Weil Conjectures.

A note on AI assistance This book has been in development since 2020. Since sometime in 2026 I have used Claude Sonnet 5 to assist with editing, checking, and writing exposition, and to port the book from LaTeX to Typst in October 2026. The overall architecture and goals of the book remain the same and are my own. Where a proof itself was drafted by Claude rather than by me, this is disclosed explicitly at the start of that proof. Some larger sections have most of their proofs drafted in this way, notably those on formal power series and Laurent series, abelian categories and categorical limits in the foundations, sheaves, and the sections developing the residue theorem. And where necessary I've prompted, or manually edited, the proofs into something more understandable. Ultimately I intend to replace these Claude-drafted proofs with my own, more idiomatic (and less pedantic) versions as time allows.

This pulls together many sources (standard or otherwise) in order to create a self-contained presentation with almost no prerequisites. At various points I have had to interpolate between an abstract, conceptual approach and a more concrete, down-to-earth approach. Many choices have been towards a more concrete approach (for example, focus on algebraic varieties rather than schemes, concrete construction of sheafification and a more explicit form of Krull's Hauptidealsatz) where I feel the abstract approach is not particularly more enlightening and can anyway be easily derived from the more concrete approach.

An interesting example is the theory of Algebraic Function Fields of One Variable / Regular Algebraic Curves, where I have interpolated between several classical treatments while departing from all of them in a number of respects – not yet ported to this edition.

On the other hand, various approaches to simpler matters have taken a more conceptual approach (for example, Krull Dimension, Dimension Theory of Field Extensions, Multilinear Algebra, Theory of Riemann Surfaces, extensive use of category-theoretic framing of various constructions such as tensor products).

For the section on Algebraic Geometry I've tried to interpolate a little between a more elementary approach (e.g. [Ser55], [Mil17]) and a scheme-theoretic approach [Stacks] [Gro64]. The idea is we may work over non-algebraically closed fields without the full (tedious) machinery of schemes. In these notes the proofs are technically simpler than the corresponding scheme results but also convey the key ideas used in the fully general case.

Similarly to [Stacks], I have made extensive use of cross-referencing so that logical chains of reasoning can be more easily followed. This naturally means many of the proofs are not as self-contained as might otherwise be the case. On the other hand we do obtain some "economy of scale", for example

- Commutative Algebra, in particular the theory of affine algebras, is the base for Algebraic Varieties and function fields
- Field Theory (separability, transcendence degree, normal extensions) is applied to étale algebras, base change and function fields, and so to varieties over non-algebraically closed fields
- Matroid theory may be used for both vector spaces and finitely generated field extensions
- "Krull Dimension" may be abstracted to work for both Rings and Topological Spaces / Varieties
- Dedekind Domain results are used for both algebraic function fields and regular algebraic curves, and the same theory would apply directly to algebraic number fields
- Theory of Representable and Adjoint Functors may be applied fruitfully to many cases, to for example tensor products and sheaves
- The language of sheaves and spaces of functions is used for both algebraic varieties and classical Riemann surfaces

I also list a non-exhaustive set of references I found useful.

Set Theory, Lattices, Category Theory

- On the Structure of Abstract Algebras [Bir35]

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- Lattice Theory [Bir40]
- A Course in Universal Algebra [BS81]
- Algebraic Theories [ARV10]
- Abelian Categories [Fre64]
- Handbook of Categorical Algebra 2: Categories and Structures [Bor94]
- Categories for the Working Mathematician [Lan98]
- Algebra: Volume I [Wae91]

Algebra

- Algebra [Lan11]
- Introduction to Commutative Algebra [AM69]
- Commutative Algebra [Mat70]
- Commutative Rings [Kap74]
- Steps in Commutative Algebra [Sha00]
- Commutative Algebra I [ZS75]
- Algebra I: Chapters 1–3 [Bou98a]
- Algebra: Chapters 4–7 [Bou89]
- Algebraic Number Theory [Neu99]
- Local Fields [Ser79]
- Residues of Differentials on Curves [Tat68]
- Stacks Project [Stacks]

Algebraic Geometry

- Algebraic Geometry [Har13]
- Algebraic Geometry [Mil17]
- Algebraic Geometry and Arithmetic Curves [Liu06]
- Algebraic Curves [Ful08]
- The Red Book of Varieties and Schemes [Mum99]
- Faisceaux algébriques cohérents [Ser55]
- Éléments de géométrie algébrique IV [Gro64]

Algebraic Function Fields

- Introduction to the Theory of Algebraic Functions of One Variable [Che51]
- Groupes algébriques et corps de classes [Ser59]
- Algebraic Function Fields and Codes [Sti09]
- Algebraic Geometry (Riemann surfaces and function fields) [Bum98]
- Introduction to Algebraic and Abelian Functions [Lan82]

Topology, Analysis and Geometry

- General Topology [Kel71]
- General Topology [Wil70]
- Commutative Algebra: Chapters 1–7 [Bou98b]
- Some remarks on biequidimensionality of topological spaces and Noetherian schemes [Hei17]
- Differential Calculus [Car71]
- Foundations of Differentiable Manifolds and Lie Groups [War13]
- Manifolds, Sheaves, and Cohomology [Wed16]
- Real and Functional Analysis [Lan12]

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